

Jialong Li (李嘉龙)

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Birthday: 1994/01

Address: Room 636, Building 19A, Shenzhen University of Advanced Technology, Shenzhen, China

Research Interests

**Computer Networks, Data Center Networks, Optical Networks,
Optical Communication, Network for Machine Learning, Network System**

Education

Tsinghua University (清华大学) Beijing, China
Doctor of Philosophy, Electronic Engineering 07/2016 – 10/2021
Thesis: Research on the convergence of optical access and metro networks
Advisor: [Bingkun Zhou](#), [Xiaoping Zheng](#)

Tsinghua University (清华大学) Beijing, China
Bachelor of Engineering, Electronic Engineering 07/2012 – 06/2016

Professional Experience

Shenzhen University of Advanced Technology (深圳理工大学) China
Assistant Professor 05/2025 – Present

Max Planck Institute for Informatics (MPI-INF) (德国马普所) Germany
Postdoctoral Associate 11/2021 – 05/2025
Advisor: [Yiting Xia](#), [Anja Feldmann](#)

Publications

[1] [Uniform-Cost Multi-Path Routing for Reconfigurable Data Center Networks](#)
Jialong Li, H. Gong, F. De Marchi, Y. Lei, A. Gong, W. Bai, Y. Xia
ACM SIGCOMM 2024, CCF-A

[2] [Matryoshka: Realizing Hyperscale Data Center Network Design for the AI Era](#)
Y. Cai*, **Jialong Li***, K. Akpinar, P. Li, H. Morsy, J. Wilson, S. Khaunte, Y. Xia, Y. Zhang
*NSDI 2026, CCF-A (*Equal Contribution)*

[3] [Optimizing Mixture-of-Experts Inference Time Combining Model Deployment and Communication Scheduling](#)

Jialong Li, S. Tripathi, L. Rastogi, Y. Lei, R. Pan, Y. Xia
IEEE/ACM Transactions on Networking (ToN, CCF-A), 2026

[4] [Unlocking Diversity of Fast-Switched Optical Data Center Networks with Unified Routing](#)

Jialong Li, F. De Marchi, Y. Lei, R. Joshi, B. Chandrasekaran, Y. Xia
IEEE/ACM Transactions on Networking (ToN, CCF-A), 2025

[5] RailS: Load Balancing for All-to-All Communication in Distributed Mixture-of-Experts Training

H. Xu, Z. Yu, C. Du, Y. Zhou, L. Li, H. Wang, W. Cheng, **Jialong Li**
IEEE/ACM Transactions on Networking (ToN, CCF-A), 2026

[6] Unlocking Superior Performance in Reconfigurable Data Center Networks with Credit-Based Transport

F. De Marchi, **Jialong Li**, Y. Zhang, W. Bai, Y. Xia
ACM SIGCOMM 2025, CCF-A

[7] Nanosecond Precision Time Synchronization for Optical Data Center Networks

Y. Lei, **Jialong Li**, Z. Liu, R. Joshi, Y. Xia
NSDI 2026, CCF-A

[8] Agora: An Open Research Framework for Optical Data Center Networks

Y. Lei, F. De Marchi, **Jialong Li**, R. Joshi, S. Wang, X. Chen, B. Chandrasekaran, Y. Xia
NSDI 2026, CCF-A

[9] FedSA-CE: Selection-Aware Clustering for Personalized Federated Learning in Heterogeneous Vehicular Networks

Y. Zhou, Z. Chen, H. Ai, H. Xu, C. Du, G. He, **Jialong Li**
IEEE Transactions on Consumer Electronics (TCE), 2026

[10] REACH: Reinforcement Learning for Efficient Allocation in Community and Heterogeneous Networks

Z. Yu, C. Du, H. Xu, Y. Zhou, B. Liu, **Jialong Li**
IEEE Transactions on Network and Service Management (TNSM), 2026

[11] Leveraging Joint Allocation of Multidimensional Resources for Distributed Task Assignment

Jialong Li, N. Hua, K. Zhu, C. Zhao, G. Pan, Y. Li, X. Zheng, B. Zhou
IEEE/Optica Journal of Optical Communications and Networking (JOCN, JCR Q1), 2022

[12] Flexible Low-Latency Metro-Access Converged Network Architecture Based on Optical Time Slice Switching

Jialong Li, N. Hua, Z. Zhong, Y. Yu, X. Zheng, B. Zhou
IEEE/Optica Journal of Optical Communications and Networking (JOCN, JCR Q1), 2019

[13] FMD-AL: Cold-Start Active Learning based on Foundation Model

H. Ai, Y. Zhou, Z. Peng, Z. Chen, S. Guo, H. Xu, Z. Yu, **Jialong Li**
ACM International Conference on Multimedia Retrieval (ICMR, CCF-B), 2026

[14] LDCS-Net: Local Dual-Context Attention Network for 3D Semantic Segmentation

S. Guo, Y. Zhou, M. Song, H. Ai, **Jialong Li**, Z. Chen

[15] A Flexible Low-Latency Metro-Access Converged Network Approach Based on Time-Synchronized TWDM-PON

Jialong Li, N. Hua, Y. Yu, Z. Zhong, X. Zheng, B. Zhou
Optical Fiber Communications Conference (OFC), 2018

[16] Balancing Energy Efficiency and Device Lifetime in TWDM-PON under Traffic Fluctuations

Jialong Li, Z. Zhong, N. Hua, X. Zheng, B. Zhou
IEEE Communications Letters (CL, JCR Q2), 2017

[17] Towards Low-Latency Distributed Tasks Collaboration by Joint Optimization of Transmission, Computation and Storage Resources Allocation in Edge Computing

Jialong Li, N. Hua, C. Zhao, Y. Li, X. Zheng, B. Zhou
Asia Communications and Photonics Conference (ACP), 2020

[18] Joint Optimization of Multidimensional Resources Allocation in Cloud Networking

Jialong Li, K. Zhu, N. Hua, C. Zhao, Y. Li, X. Zheng, B. Zhou
IEEE 7th Optoelectronics Global Conference (OGC), 2022

[19] Hop-On Hop-Off Routing: A Fast Tour across the Optical Data Center Network for Latency-Sensitive Flows

Jialong Li, Y. Lei, F. De Marchi, R. Joshi, B. Chandrasekaran, Y. Xia
6th Asia-Pacific Workshop on Networking (APNet 2022)

[20] Generalized SRv6 Header Compression Packet Processors based on Multicore Architectures

W. Cheng, H. Wang, X. Duan, H. Li, **Jialong Li**, Z. Cheng
Poster, APNet 2025

[21] Rethinking Transport Protocols for Reconfigurable Data Centers: An Empirical Study

F. De Marchi, **Jialong Li**, W. Bai, Y. Xia
ACM SIGCOMM 2024 Workshop on Hot Topics in Optical Technologies and Applications in Networking (HotOptics 2024)

[22] TopoFair: A Fairness-Aware Topology Selection for Reconfigurable Data Center Networks

Z. Yu, C. Du, H. Xu, H. Wang, B. Liu, **Jialong Li**
Asia Communications and Photonics Conference and Exhibition (ACP 2025), Oral

[23] [Learning Optimal Transport for GPU Allocation: A Deep Learning Approach to Efficient Resource Scheduling](#)

C. Du, Z. Yu, H. Xu, H. Wang, B. Liu, **Jialong Li**

Asia Communications and Photonics Conference and Exhibition (ACP 2025), Oral

[24] [Efficient Flow Scheduling in Distributed Deep Learning Training with Echelon Formation](#)

R. Pan*, Y. Lei*, **Jialong Li**, Z. Xie, B. Yuan, Y. Xia

Twenty-First ACM Workshop on Hot Topics in Networks (HotNets 2022)

[25] [Provisioning Short-Term Traffic Fluctuations in Elastic Optical Networks](#)

Z. Zhong, N. Hua, M. Tornatore, **Jialong Li**, Y. Li, X. Zheng, B. Mukherjee

IEEE/ACM Transactions on Networking (ToN, CCF-A), 2019

[26] [Opportunistic Credit-Based Transport for Reconfigurable Data Center Networks with Tidal \(Third Place\)](#)

F. De Marchi, **Jialong Li**, W. Bai, Y. Xia

Poster, ACM SIGCOMM 2024

[27] [An Open Research Framework for Optical Data Center Networks](#)

Y. Lei, R. Joshi, F. De Marchi, **Jialong Li**, B. Chandrasekaran, Y. Xia

Demo, ACM SIGCOMM 2024

[28] [Nanosecond Precision Time Synchronization for Optical Data Center Networks](#)

Y. Lei, **Jialong Li**, Z. Liu, R. Joshi, Y. Xia

Poster, ACM SIGCOMM 2025

[29] [Provisioning Uninterrupted Satellite Communication Services by Preset-Satellite-Chain \(PSC\)-Based Seamless Handover \(Best Poster Award\)](#)

C. Zhao, N. Hua, **Jialong Li**, X. Zheng

Asia Communications and Photonics Conference (ACP), 2020

[30] [Achieving Ultralow-Latency Optical Interconnection for High Performance Computing \(HPC\) Systems by Joint Allocation of Computation and Communication Resources \(Top-Scored Paper\)](#)

R. Luo, Y. Yu, N. Hua, Z. Zhong, **Jialong Li**, X. Zheng, B. Zhou

Optical Fiber Communications Conference (OFC), 2019

[31] [Time-Sliced Flexible Resource Allocation for Optical Low Earth Orbit Satellite Networks](#)

Z. Zheng, N. Hua, Z. Zhong, **Jialong Li**, Y. Li, X. Zheng

IEEE Access, 2019

[32] Achieving Global Optical Spectral Perception via Unified Information Model and Adaptive Perceptual Information Calibration

K. Zhu, N. Hua, C. Zhao, Y. Li, **Jialong Li**, Y. Li, X. Zheng, B. Zhou
Optoelectronics and Communications Conference (OECC), 2021

[33] Throughput Scaling for MMF-Enabled Optical Datacenter Networks by Time-Slicing-based Crosstalk Mitigation

Z. Zhong, N. Hua, Y. Yu, Z. Wu, J. Li, H. Yan, S. Li, R. Luo, **Jialong Li**, Y. Li, X. Zheng
Optical Fiber Communications Conference (OFC), 2018

[34] Fast-Reconfigurable Optical Interconnect Architecture based on Time-Synchronized Node Coordination for High Performance Computing

Y. Yu, N. Hua, Z. Zhong, **Jialong Li**, R. Luo, Z. Zheng, X. Zheng
Asia Communications and Photonics Conference (ACP), 2017

Teaching

Max Planck Institute for Informatics (MPI-INF)

Co-Lecturer, Data Networks

Spring 2022

Tutor, Hot Topics in Data Networks Seminar

Fall 2021

Tutor, Hot Topics in Data Networks Seminar

Fall 2022

Tsinghua University

TA, Introduction to Information Science and Technology

Fall 2018

TA, Introduction to Information Science and Technology

Fall 2017

Professional Service

Conferences

Program Committee, APNet 2026

Program Committee, ACM IMC 2026

Program Committee, ACM SIGCOMM 2024 Workshop on HotOptics

Program Committee, ICNT 2024

Program Committee, 24th Asia Communications and Photonics Conference (ACP/IPOC 2024)

External Reviewer, NSDI 2026

Reviewer, WCSP 2019

Journals

Guest Editor, Photonics

Reviewer, IEEE/ACM Transactions on Networking

Reviewer, IEEE/Optica Journal of Optical Communications and Networking

Reviewer, IEEE Transactions on Communications

Reviewer, IEEE Transactions on Wireless Communications

Reviewer, Optics Express

Reviewer, IEEE Internet of Things Journal

Reviewer, IEEE Communications Letters

Reviewer, Journal of Cybersecurity and Privacy

Reviewer, Mathematics
Reviewer, Photonics
Reviewer, Applied Science
Reviewer, Electronics
Reviewer, Sensors
Reviewer, Computer Networks and Communications

Academic Service

Web Chair, ACM SIGCOMM 2024 Workshop on HotOptics, 2024
Committee member of Max Planck Institute Internship Program, 2022
Student Chairman of Beijing Optical Society Tsinghua University Chapter,
 2018 – 2019

Awards and Honors

Tsinghua's Friend – Xiaomi Corporation Scholarship (Tsinghua University)	2020
The Second Prize of Comprehensive Scholarship (Tsinghua University)	2018
Excellence Scholarship for Social Work (Tsinghua University)	2014

Student Mentoring

Master Thesis

Vadim Farutin, Saarland University, 10/2022 – 08/2023, Topic: Network-Aware GPU Sharing for Distributed Deep Learning, Score: 1.0 (**Highst**)

Interns

Rui Pan, University of Wisconsin-Madison, 02/2022 – 08/2022, Now Phd student@Princeton
Haotian Gong, The University of British Columbia, 05/2023 – 08/2023, , Now Phd student@University of Michigan
Aoyu Gong, EPFL, 07/2023 – 02/2024, Now Phd student@EPFL
Max Huang, University of Waterloo, 01/2024 – 04/2024
Laskhay Rastogi, Indian Institute of Technology (IIT), Kanpur, 10/2023 – 01/2024
Shreyansh Tripathi, Saarland University, 05/2024 – Present

Selected Talks

Uniform-Cost Multi-Path Routing for Reconfigurable Data Center Networks
 AI-Native Infrastructure: Foundations and Algorithmic Frontiers, SIAT, Shenzhen, China, 10/2025
 ACM SIGCOMM 2024, Sydney, Australia, 08/2024
 AI Time on Networking Seminar, online, 10/2024
 Asia Communications and Photonics Conference Workshop, Beijing, China, 11/2024

Routing and Scheduling in Optical Data Center Networks for Emerging Cloud Applications
 Shanghai Jiao Tong University, Shanghai, China, 02/2024
 Fudan University, Shanghai, China, 02/2024
 ShanghaiTech University, Shanghai, China, 02/2024
 University of Science and Technology of China, Hefei, China, 03/2024

**Hop-On Hop-Off Routing: A Fast Tour across the Optical Data Center Network
for Latency-Sensitive Flows**

6th Asia-Pacific Workshop on Networking (APNet 2022), online, 07/2022

Tsinghua-Berkeley Shenzhen Institute, Shenzhen, China, 04/2023

**Leveraging Joint Allocation of Multidimensional Resources for Distributed
Task Assignment**

IEEE 7th Optoelectronics Global Conference (OGC 2022), online, 12/2022

Conference on Information Optics and Photonics, Xi'An, China, 07/2021

**Building Low-Latency and Energy-Efficient Optical Metro-Access Converged
Networks**

Max Planck Institute for Informatics, online, 12/2020

University of Vienna, online, 04/2021

Updated on 04/2026